

THE MECHANISM OF THE RHYTHMIC CHANGES
IN THE CALIBRE OF THE BRONCHI
DURING RESPIRATION

By MAXWELL ELLIS¹

(From the Medical Unit, University College Hospital, London)

(Received April 16, 1936.)

IN his description of the bronchial tree Reisseisen [1822] concluded that the structure was designed to allow of longitudinal movements of the bronchi during the inspiratory phase of respiration. This hypothesis has since been confirmed and amplified by many workers using many different methods. In recent years it has been studied by means of radiographic delineation of radio-opaque media instilled into the bronchial tree [Bullock and Gottlieb, 1920, 1922], and the movement has actually been seen through a bronchoscope [Ingals, 1905; Jackson, 1917], as well as measured by Weingaertner [1920]. Such a longitudinal change in the bronchus is readily appreciated as a mechanical result of stretching when the shrunken lung expands with the inspiratory expansion of the thorax. Keith [1909] showed how this is achieved in the upper lobe by the attachment of the crura of the diaphragm to the lung root, and the consequent descent of this point with contraction of the diaphragm. Until recently there have been no means of determining what other bronchial changes occur.

If the bronchus is considered as an elastic tube, elongation would be accompanied by narrowing of the lumen, but Ingals and Jackson saw widening of the bronchus during inspiratory movements, both in animals and man, and this phenomena, as seen through the bronchoscope, has been confirmed by many observers. Radiographic results are inconclusive, as the smallness of the changes involved makes recognition of them uncertain. A method recently described for the direct recording of changes in calibre [Ellis and Livingston, 1935] clearly demonstrates that the bronchial tubes widen in the inspiratory phase of respiration and

¹ Geoffrey E. Duveen Student in Oto-Rhino-Laryngology, University of London.

narrow down during the expiratory phase. These alterations in calibre are synchronous with the respiratory movements, and always accompany them, and the present experimental enquiry was conducted to elucidate their mechanism.

There are several possible sources for the production of these changes in calibre:

- (1) An intrinsic rhythm of the smooth muscle in the walls of the tubes.
- (2) The result of nervous impulses carried in the vagus or sympathetic nerves.
- (3) A mechanical result of chest wall movements, the dilatation being purely passive.

Trendelenburg [1912] investigated the behaviour of isolated strips of bronchial muscle in the ox and sheep, and came to the conclusion that it possessed no power of spontaneous contractile or expansile movement. The second hypothesis has been advanced in Starling's *Principles of Physiology* [1933], where it is stated that possibly nervous impulses are responsible for active contraction of the bronchial muscle in the expiratory phase. Some indirect support is lent to this theory by the existence of sensory spindles in the muscle layer of the bronchial walls [Larsell, 1923; Gaylor, 1934], and by the presence of sensory fibres leaving the lungs in which action currents can be detected at each inspiratory movement [Ranson, 1921; Adrian, 1933]. This current resembles the discharge from stretch receptors in voluntary muscle, and this may be one side of a reflex arc of which the complement is active contraction of the bronchial muscle.

EXPERIMENTAL METHOD

Experiments on dogs were directed along three lines:

- (1) To imitate the changes in calibre mechanically.
- (2) To observe the effect of nerve section on the movements.
- (3) To observe the movements from a bronchus in a lung which is rendered motionless by resection of a large portion of the thorax while the other lung is functioning and the nerve connexions are at first intact, and subsequently divided.

For the first type of experiment the principle of the Drinker respirator was utilized. The trachea and lungs were removed from the animal with care to avoid injuring the visceral pleura. A glass tube was tied tightly into the trachea, and the whole preparation placed in a bell-jar. Through the rubber stopper of this jar passed the glass tube from the trachea and another tube which projected an inch or so on either side, fitted on the outer side with a stopcock. A sheet of rubber was fastened

to the base of the bell-jar (Fig. 1). A partial vacuum was created in the jar and the lungs thus made to expand. A recording instrument (see below) was inserted through the tube in the trachea into the right bronchus and the connections made. Pressure on the rubber bottom of the jar caused contraction of the lungs, and respiratory movements could thus be stimulated.

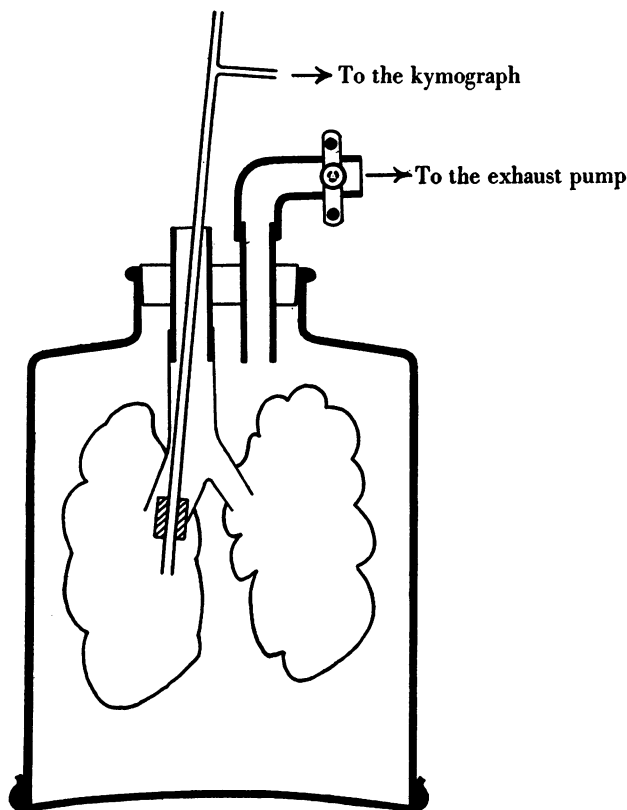


Fig. 1. A diagram to show how respiratory movements were simulated in the isolated lungs.

The second and third series of experiments involved the use of an instrument carrying a balloon which was inserted through a specially made bronchoscope into the right main bronchus of a dog. A full description of the instrument and recording apparatus is given in a previous paper [Ellis and Livingston, 1935]. When the instrument is in position, the balloon can be inflated. There is a hollow tube traversing the balloon and projecting beyond it so that the respiratory exchange can continue in the portion of the lung distal to the balloon. The external

diameter of the instrument is 4.0 mm. The balloon is connected to a small Brodie bellows which records its changes in volume. These are due to local compression or relaxation of the narrowing or widening bronchus. Dogs of between 7.0 and 9.0 kg. were found to be suitable. Anæsthetization was sometimes carried out with subcutaneous morphine sulphate and intraperitoneal sodium amytal, and sometimes with intravenous chloralose (0.1 g. per kg. body weight). A few animals were decerebrated. Seventy-two animals in all were used.

Division of the spinal cord had to be accomplished whilst the animal was firmly fastened, supine, with records being obtained from the bronchus and chest wall. A preliminary laminectomy or exposure of the occipito-atlantal space was, therefore, performed, and the prepared area placed over an opening in the table. The cord could thus be divided without disturbing the animal's position or the recording apparatus. The vagal nerves were isolated in the neck in the usual way.

In the third group of experiments, with the object of immobilizing the right lung, an artificial pneumothorax was sometimes induced, by injection of air, and, sometimes, a large portion of the thoracic wall on the right side was resected. This latter was performed almost bloodlessly by ligature of the ribs along the sternal margin and in the axilla, and resection of the intervening piece of the chest. The animal was kept alive after the thorax was opened by artificial respiration of the left lung. This was achieved by passing an instrument similar to the one in the right bronchus down to the left bronchus through an opening in the trachea, connecting its central lumen with an artificial respiration pump, and inflating the balloon so that no air could be blown back past it and enter the right bronchus. When this operation was completed, the right lung was exposed to view and seen to be collapsed and motionless. The heart was clearly visible, and the inflation and deflation of the left lung could easily be seen through the thin mediastinum. This movement of the left lung caused a shifting over of the mediastinal contents at each inflation, and this sometimes was sufficiently marked to cause a compression of the recording balloon in the right bronchus.

The experiment usually took the form of first obtaining a normal record and testing the sensitivity of the bronchial preparation with one or two small injections of adrenaline, which caused temporary relaxation of bronchial tone. The chest wall was then resected, and, later, the vagi were divided one after the other. The peripheral ends were often stimulated with weak faradic currents, and, finally, the spinal cord was divided through the occipito-atlantal space.

RESULTS

The first group of experiments showed that it is possible to produce mechanically a record (Fig. 2) identical with the usual record from an intact animal, and that the changes in calibre occurred synchronously with the artificially induced expansion and contraction of the lungs.

The information from the second group was that with bilateral vagotomy the alterations in calibre are intensified. In spite of the removal of broncho-constrictor influences the bronchi narrowed, in the expiratory phase, to a greater extent than before (Fig. 3), and also opened more widely during the inspiratory phase. Section of the spinal cord in the lower cervical region, removing sympathetic broncho-dilator control, also

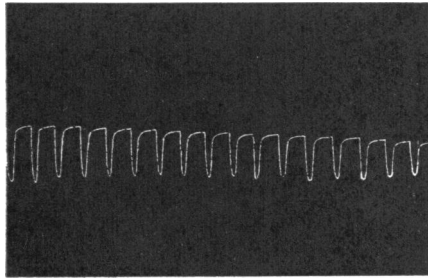


Fig. 2. A record from the right bronchus of the isolated lung preparation of Fig. 1.

paralysed all respiratory movements, but, when artificial respiration was given, the bronchus opened and closed precisely as before. The rhythm seen was always an exact counterpart, even to every chance momentary variation, of that recorded for the respiratory movements of the chest wall. Evidently, impulses to produce this rhythmic movement are carried neither in the sympathetic nor in the vagal nerves.

The third group of experiments was carried out in two ways. The first merely consisted in inducing a right-sided artificial pneumothorax, in order to render the corresponding lung as free as possible from the influence of thoracic movements. The resulting record (Fig. 4) differs but little from a normal record. However, the lung is not completely immobilized in these conditions, and reflects to a considerable extent the movements of the thorax. To effect such an immobilization a large portion of the right thoracic wall was removed (see above). Owing to the porous nature of the dog's mediastinum this would rapidly create a bilateral

pneumothorax, and the left lung was, therefore, kept in motion by the technique already described.

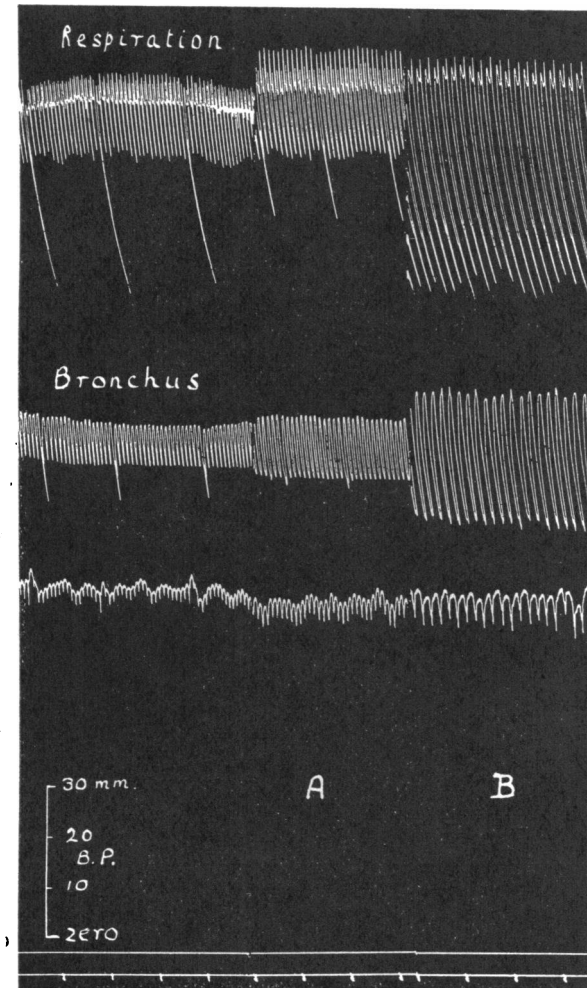


Fig. 3. Dog. Records from the right bronchus after right (A) and left (B) vagotomy, showing the increase in amplitude of the bronchial movements. Respiratory movements are recorded by a pneumatic stethograph from the chest wall.

The bronchial movements were successively recorded as follows:

- A. In the intact animal.
- B. After removal of the thoracic wall and bilateral phrenicectomy.

C. After right and then bilateral vagotomy.

D. And, finally, after division of the spinal cord at the level of the occipito-atlantal space.

It will be seen that each operation reduced the amount of movement undergone by the thoracic cage, until it was completely motionless. The

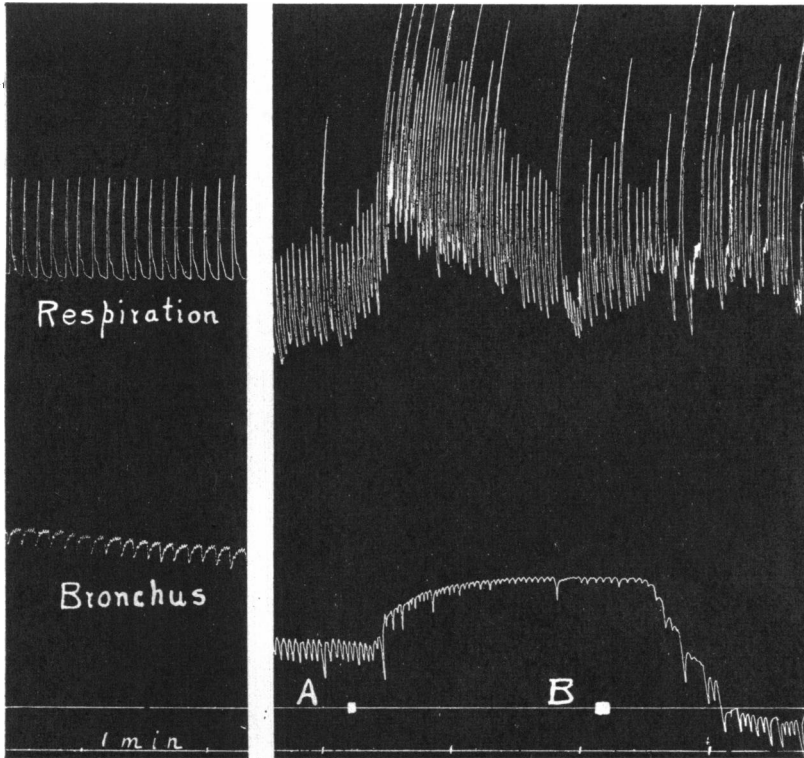


Fig. 4. Dog. Records from the right bronchus before and after the induction of a right pneumothorax of moderate degree. There is very little alteration in the amplitude of the movements. The reactions to pilocarpine (at A) and atropine (at B) during the pneumothorax are as in the normal animal.

left lung in the final stage is inflated at each stroke of the pump, but this causes only very slight distension of the left side of the thorax.

The results obtained were consistent. Fig. 5 is a record obtained from an intact animal. Stage B is illustrated by Fig. 6. The respiratory rate is synchronous with the pump strokes, and the animal with intact vagi always takes up this rhythm sooner or later, whatever its previous

rate might have been. After bilateral vagotomy (Fig. 7) a respiratory rhythm independent of the pump strokes becomes established, and that of the bronchi is synchronous with the new rate, but there is no alteration in the amplitude of the excursion. When stage D is reached, the rhythm of the very small excursions seen is again synchronous with the speed of

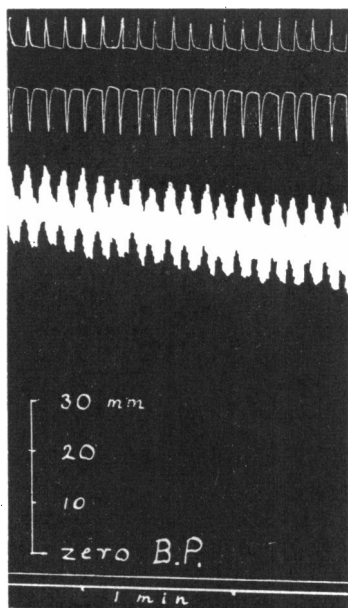


Fig. 5.

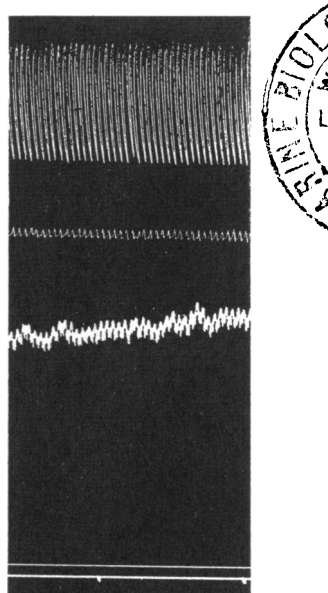


Fig. 6.

Fig. 5. Dog. (In this and all the following figures the records from above downwards are of respiration, bronchial movements, blood pressure, signal line and time marker.) Curves from an intact animal.

Fig. 6. This and the following figure are from the same animal as in Fig. 5. It shows the reduction in amplitude of the bronchial movements following resection of the chest wall and bilateral phrenicectomy, and the new rhythm, that of the pump, taken up synchronously by both the chest and bronchus.

the pump (Fig. 8, taken from another experiment where inflation of the left lung was greater owing to the smaller size of the dog), and an identical record is obtained after the animal is killed with an intravenous injection of chloroform. The upstrokes and downstrokes in this last record indicate that it is produced by mediastinal pressure on the balloon with inflation of the left lung, which direct observation confirmed.

In a few experiments the procedure was slightly altered. The vagi and spinal cord were divided before resection of the thoracic wall was per-

formed. After division of the cord the thorax becomes almost motionless—artificial inflation of the left lung (carried out as described above) being only just sufficient to cause a slight movement of the left side. Very small waves were recorded from the right bronchus synchronous with the pump rate and the upstroke corresponding with inflation; the curve remained

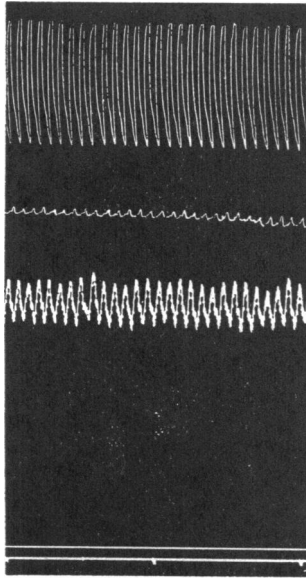


Fig. 7.

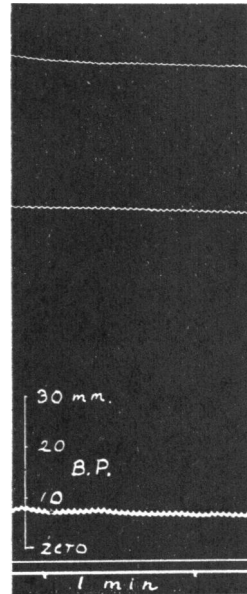


Fig. 8.

Fig. 7. Bilateral vagotomy has been followed by a different rhythm, in spite of the fact that the pump is still inflating the left lung at the same rate as before, but the amplitude of the bronchial movements remains unchanged.

Fig. 8. Dog. After division of the spinal cord the left lung, inflated by the pump, causes very slight distension of the chest. A record from another animal shows that a slight movement sometimes occurs in these conditions. This record is unchanged by the death of the animal, and is mechanically produced by mediastinal shift.

unaltered when the chest was laid open, and also when the animal was killed. It was seen that the excursions were produced by mediastinal shift in the manner already described. Admittedly, in the final phases of the experiments the blood pressure is low, and the condition of the animal not favourable for the production of a spontaneous smooth muscle rhythm, but, in any case, no sign of such a rhythm ever appeared.

DISCUSSION

In the preamble the existence of rhythmic bronchial movements synchronous with respiratory movements was pointed out, and three possible means of production of these movements were indicated. The experiments described show that these movements can be produced mechanically. Also, they occur in the denervated bronchus if the thoracic wall is kept moving, and are synchronous with this movement.

The preparation in stage B of the third series of experiments consisted of intact nerves, a functioning left lung, and a sensitive recording instrument in the right bronchus. If impulses for reflex contraction of the bronchial muscle were being conveyed from the left bronchus, they were not prevented from exercising their effect on the right bronchus, and any such effect would become obvious from the record. The second possible means of production was thus directly tested, but with negative results, as no movements whatever were recorded from the right side when the thorax was motionless.

Moreover, it was seen that reduction of thoracic movements reduced bronchial movements, and it can only be concluded that the latter depend upon and are a direct result of the former. The reflex arc, the sensory side of which starts from stretch receptors in the bronchial muscles, is thus not concerned with rhythmic contraction of bronchial muscle. It may assist in controlling the respiratory movements of the thorax. There appears to be no rhythmic function assignable to the bronchial muscle, which seems to be concerned entirely with maintaining a moderate tone, and hence a lumen in the smaller bronchi which might otherwise be completely occluded during expiratory movements. In Fig. 9, for example, taken from an animal after resection of the chest wall, a distinct relaxation of the bronchus is seen when the right vagus is divided, because of removal of the tonic impulses conveyed along the broncho-constrictor nerves. Fig. 10 shows the effect of stimulation of the peripheral end of the cut vagus in another animal, after resection of the chest wall, and demonstrates clearly that the bronchus is wide open and relaxed. A similar effect can be produced in similar circumstances by the administration of pilocarpine. It occurs even when the œsophagus is immobilized by being filled with plaster of Paris, and is thus not due to the tetanic contraction of the œsophageal muscle [Ellis and Livingston, 1935] exerting an effect from without.

The work described found no evidence of a rhythmic alteration in calibre as a property of bronchial muscle. Such movements do occur,

and are synchronous with the respiratory movements of the thoracic cage, but they are due entirely to the mechanical effect of the latter.

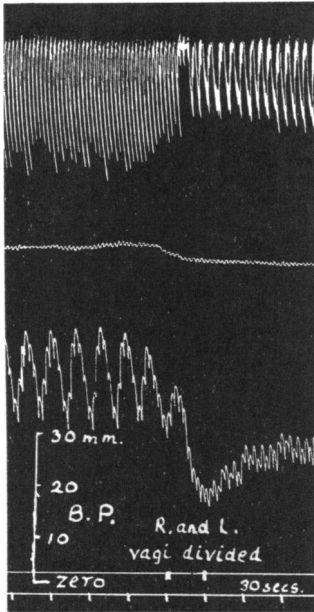


Fig. 9.

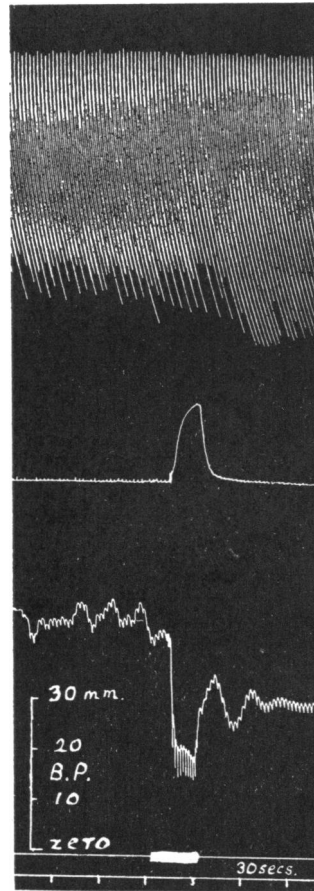


Fig. 10.

Fig. 9. Dog. A record showing the relaxation in tone of the bronchus after division of the right vagus—the chest wall has been previously resected.

Fig. 10. Dog. A record obtained in similar circumstances to that in Fig. 9, after right-sided vagotomy, showing the increase in bronchial tone when the peripheral end of the cut vagus is stimulated with a faradic current.

Bronchial muscle possesses the function of maintenance of tone. This conclusion is in agreement with its anatomical structure and its character—smooth muscle. In normal conditions, this tone has been shown to be constantly present—it can be lowered by the administration of adrenaline,

and is controlled by the sympathetic and vagus nerves. (Occasionally, adrenaline has been noted to give rise to an increase of tone, observed also by Kiese [1935], but this may be a secondary effect of the rise of blood pressure acting centrally in an unusual manner.)

SUMMARY

1. The nature of the rhythmic alterations in calibre of the bronchi seen during bronchoscopy is analysed by a direct instrumental method.

2. The rhythm of these movements under all conditions of experiment in the dog is identical with that of the respiratory movements of the chest wall.

3. It is changed by section of the vagi, of the phrenic nerves, or of the cervical spinal cord only as far as the respiratory movements of the chest wall are thereby affected.

4. By rendering the lung motionless the rhythm is abolished.

5. It is concluded that all rhythmic changes in calibre are produced passively by the respiratory movements of the thorax.

6. Alterations in tone are demonstrated, and it is held that the chief function of the bronchial muscle is the maintenance of tone.

REFERENCES

- Adrian, E. D. (1933). *J. Physiol.* **79**, 332.
- Bullock, J. G. M. and Gottlieb, C. (1920). *Amer. J. med. Sci.* **160**, 98.
- Bullock, J. G. M. and Gottlieb, C. (1922). *Laryngoscope*, **32**, 284.
- Ellis, M. P. and Livingston, A. E. (1935). *J. Physiol.* **84**, 223.
- Gaylor, J. B. (1934). *Brain*, **57**, 143.
- Ingals, E. F. (1905). *J. Amer. med. Assoc.* **45**, 1302.
- Jackson, C. (1917). *Ibid.* **68**, 245.
- Keith, A. (1909). *Further Advances in Physiology* (ed. Leonard Hill). London.
- Kiese, M. (1935). *Arch. exp. Path. Pharmac.* **178**, 342.
- Larsell, O. (1923). *J. comp. Neurol.* **35**, 97.
- Ranson, S. W. (1921). *Physiol. Rev.* **1**, 477.
- Reisseisen, F. D. (1822). *Ueber den Bau der Lungen*. Berlin.
- Starling, E. H. (1933). *Principles of Physiology*, p. 912. London.
- Trendelenburg, P. (1912). *Arch. exp. Path. Pharmac.* **69**, 79.
- Weingaertner, M. (1920). *Arch. Laryngol. Rhin., Berl.* **32**, 1.